

# RASSIL BEN RHOUMA

CTO & Co-Founder — Laevitas

Tunis, Tunisia | +216 52 497 659 | br.rassil@gmail.com | rassilbr.me

## PROFILE

CTO and co-founder of **Laevitas**, a crypto derivatives analytics platform used by trading desks and quant teams. I architect and run the full stack: event-driven ingestion on **NATS JetStream** and Redis Streams, a **ClickHouse** warehouse on bare metal holding 7B+ rows of tick data, and a public market-data API at **99.99% availability** on self-managed **Kubernetes**. I moved the platform off managed cloud databases onto bare metal — 92% cheaper and faster — and I own the commercial layer: Stripe subscriptions, usage quotas and on-chain pay-per-request billing. Nine years in real-time data infrastructure, still hands-on in the codebase.

## HIGHLIGHTS

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| <b>API Availability</b><br>99.99% UPTIME<br>Four nines on the public market-data API — layered caching, rate limiting, SLO alerting.                                                        | <b>Event-Driven Core</b><br>NATS JETSTREAM<br>Analytics off cron/batch onto producer-consumer streaming with replay — sub-second latency.             | <b>Cost Optimization</b><br>92% LOWER DB SPEND<br>Managed cloud databases benchmarked against ClickHouse on bare metal, then migrated. |
| <b>Kubernetes Platform</b><br>SELF-MANAGED k3s ON BARE METAL<br>Production and analytics clusters — 7B+ rows on HA ClickHouse, zero-downtime rollouts, autoscaling, infrastructure-as-code. | <b>Payments in Production</b><br>STRIPE + x402<br>Subscriptions, entitlements and invoicing tied to API tiers, plus on-chain pay-per-request billing. |                                                                                                                                        |

## EXPERIENCE

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| <b>CTO &amp; Co-Founder — Laevitas</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 08/2020 – Present · Tunis, Tunisia |
| <ul style="list-style-type: none"><li>Own technology <b>end to end</b> — architecture, infrastructure, security and the engineering team</li><li>Re-architected core analytics from cron/batch jobs to an <b>event-driven pipeline on NATS JetStream and Redis Streams</b> (producer-consumer, replay, at-least-once delivery), bringing data latency down to sub-second</li><li>Moved the data layer <b>from OLTP to OLAP</b>: after 1.5 TB of lagging market data on MongoDB Atlas, AWS Aurora and other managed databases, benchmarked and consolidated onto <b>ClickHouse on bare metal</b> — 7B+ rows, sub-second queries, and database spend down from <b>\$9,000 to \$700</b> a month</li><li><b>Architected and operate self-managed Kubernetes (k3s) on bare metal</b> across production and analytics clusters — HA ClickHouse, zero-downtime rollouts, autoscaling, infrastructure-as-code</li><li>Raised public API availability to <b>99.99%</b> with layered caching, sliding-window rate limiting, graceful degradation and Prometheus/Grafana SLO alerting on every hop</li><li>Shipped <b>AI-native access</b>: an MCP server exposing 30+ market-data tools with OAuth 2.1 so AI agents query Laevitas directly, plus <b>Sentinel</b> — LLM-driven anomaly detection and alerting over live derivatives flow</li><li>Own <b>billing and payments end to end</b>: subscription and payment logic on <b>Stripe</b> (plans, entitlements, upgrades, webhooks, invoicing) wired to API-key tiers and usage quotas, plus <b>x402</b> on-chain pay-per-request billing with prepaid credit bundles settled in ~1 ms</li><li>Built observability from zero across ingestion, storage and API (Grafana, Prometheus, Elasticsearch) and automated internal runbooks with <b>n8n</b>, including a custom Redis Streams trigger node</li></ul> |                                    |

## EARLIER EXPERIENCE

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|----------------------------------------------------------------------------------------------------------|-------------------|
| <b>Software Engineer &amp; Lead — Growapp</b> · SaaS marketing analytics (Spring Boot, Node.js, Angular) | 05/2020 – 03/2021 |
| <b>Full Stack Developer — Sure-Bid</b> · Subcontractor bidding platform (Node.js, Angular, MongoDB, AWS) | 02/2019 – 04/2020 |
| <b>JEE Developer — OTDAV</b> · Copyright tooling for a national institution (JEE, WildFly, MySQL)        | 11/2018 – 03/2019 |
| <b>Web Developer — Efrica Ecommerce</b> · Ecommerce platform for African artisans (PHP, Symfony)         | 07/2017 – 11/2018 |

## TECHNICAL SKILLS

|                             |                                                                                                     |
|-----------------------------|-----------------------------------------------------------------------------------------------------|
| <b>DATA &amp; STREAMING</b> | ClickHouse · NATS JetStream · Kafka · Redis (Streams, Pub/Sub) · MongoDB · PostgreSQL · Aurora      |
| <b>BACKEND</b>              | NestJS · Node.js · TypeScript · Python · Java / JEE · REST · GraphQL · WebSockets                   |
| <b>INFRASTRUCTURE</b>       | Kubernetes (k3s, bare metal) · Docker · Nomad · AWS · Hetzner · Cloudflare · GitHub Actions         |
| <b>OBSERVABILITY</b>        | Grafana · Prometheus · Elasticsearch · SLO design & alerting                                        |
| <b>PAYMENTS</b>             | Stripe (subscriptions, webhooks, invoicing) · x402 on-chain micropayments · usage metering & quotas |
| <b>FRONTEND &amp; AI</b>    | Angular · TradingView · ECharts · MCP (Model Context Protocol) · LLM-driven features · n8n          |

## EDUCATION

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|-------------------------------------------------------------------------------------------------------------------------|
| <b>Engineering Degree / Master's in Computer Science</b><br>Esprit School of Engineering · Tunis, Tunisia · 2017 – 2021 |
| <b>Bachelor's in Computer Science</b><br>Faculty of Sciences of Tunis · Tunis, Tunisia · 2014 – 2017                    |

## LANGUAGES

|                |              |
|----------------|--------------|
| <b>Arabic</b>  | Native       |
| <b>French</b>  | Professional |
| <b>English</b> | Professional |